

ECON 4140 Financial Econometrics

Lecture 2 Public Class Notes

Scatter Plots, Association, Covariance, Correlation & Causality

Summer 2026 | Public Lecture Format | Knowledge Points + Practice Questions

Lecture 2 moves from one-variable descriptive statistics to two-variable relationships. The key skill is not only calculation, but also professional interpretation: direction, strength, linearity, outliers, and causality.

1. Lecture Overview

Topic	Key Question
Scatterplots	What visual pattern exists between x and y?
Covariance	Do x and y move together or in opposite directions?
Correlation	How strong is the linear relationship after standardization?
Spearman Correlation	How strong is the monotonic relationship based on ranks?
Causality	Can we interpret association as a causal effect?

2. From One Variable to Two Variables

Week 1 used one variable to describe center and spread. Week 2 uses two variables, x and y, to describe association.

One Variable	Two Variables
Population: μ , σ^2 , σ	Population: σ_{xy} , ρ
Sample: $\bar{x}$, s^2 , s	Sample: s_{xy} , r
Center and spread	Association between x and y

Population data: $(x_1, y_1), (x_2, y_2), \dots, (x_N, y_N)$

Sample data: $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$

Practice 1 - Data Type

Bank of Canada inflation example uses monthly observations of inflation and an earlier overnight interest rate. Which type of data is this?

Answer: It is time-series data because observations are monthly values collected over time.

3. Scatterplots

A scatterplot is the first tool for checking the relationship between two quantitative variables.

Feature	What to Look For
Direction	Positive or negative association
Form	Linear, curved, clustered, or no clear form
Strength	Whether points follow a tight or loose pattern

Outliers	Unusual points that may influence correlation
Professional wording: "There appears to be a strong negative linear association between x and y."	

SAS Example

```
proc sgplot data=sashelp.class;
  scatter x=height y=weight / group=sex;
  ellipse x=height y=weight;
  reg x=height y=weight / clm cli;
run;
```

```
proc corr data=sashelp.class plots=matrix(histogram);
  var age height weight;
run;
```

Practice 2 - Scatterplot Interpretation

A scatterplot slopes downward from left to right and the points are close to a straight line. How should we describe the relationship?

Answer: There is a strong negative linear association between the two variables.

4. Covariance

Covariance describes whether two variables tend to move together. However, its magnitude depends on measurement units, so it is not a good measure of relationship strength.

Population Covariance

$$\sigma_{XY} = \frac{\sum (x_i - \mu_X)(y_i - \mu_Y)}{N}$$

Sample Covariance

$$s_{xy} = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{(n - 1)}$$

Covariance Sign	Interpretation
$s_{xy} > 0$	x and y tend to move in the same direction
$s_{xy} < 0$	as x increases, y tends to decrease
$s_{xy} \approx 0$	little or no linear co-movement
Important: The magnitude of covariance has no direct meaning because it depends on the units of x and y.	

Practice 3 - Interpret Covariance

Suppose $s_{xy} = -25.31$. Interpret this number professionally.

Answer: The negative sign indicates a negative association: as x increases, y tends to decrease. The magnitude 25.31 has no direct interpretation because covariance depends on measurement units.

5. Correlation

Correlation standardizes covariance and removes the effect of units. It measures the direction and strength of a linear relationship.

Population Correlation

$$\rho_{XY} = \sigma_{XY} / (\sigma_X \sigma_Y)$$

Sample Correlation

$$r_{xy} = s_{xy} / (s_x s_y) = \sum (x_i - \bar{x})(y_i - \bar{y}) / \sqrt{[\sum (x_i - \bar{x})^2 \sum (y_i - \bar{y})^2]}$$

Value of r	Interpretation
$r \approx 1$	Strong positive linear relationship
$r \approx -1$	Strong negative linear relationship
$r \approx 0$	Weak linear relationship, but not necessarily no relationship
Exam wording: Correlation near zero only means weak linear relationship. The variables may still have a nonlinear relationship.	

Practice 4 - Multiple Choice

Which statement is correct? A) Positive linear relationship means all points lie exactly on a straight line. B) Stronger positive relationship means r is closer to -1. C) Two variables that are uncorrelated may still be related nonlinearly. D) Stronger negative relationship means r is closer to +1.

Answer: Correct answer: C. Correlation only measures linear association; nonlinear relationships may have r close to zero.

Practice 5 - Correlation Interpretation

Suppose $r = -0.9421$. Interpret both sign and magnitude.

Answer: The negative sign means x and y move in opposite directions. The magnitude $|r| = 0.9421$ indicates a very strong linear relationship.

6. Spearman's Rank Correlation

Spearman correlation is the Pearson correlation applied to rank variables. It is useful when the relationship is monotonic but not necessarily linear, or when raw values contain strong outliers.

$$r_s = \rho_{R(X), R(Y)} = \text{Cov}(R(X), R(Y)) / [\sigma_{R(X)} \sigma_{R(Y)}]$$

If ranks are distinct: $r_s = 1 - [6 \sum d_i^2] / [n(n^2 - 1)]$

Pearson r	Spearman r_s
Uses raw values	Uses ranks
Measures linear association	Measures monotonic association
Sensitive to outliers	Often less sensitive to outliers

Practice 6 - Spearman Formula

For $n = 5$, suppose $\sum d_i^2 = 2$. Compute r_s .

Answer: $r_s = 1 - 6(2) / [5(5^2 - 1)] = 1 - 12/120 = 0.90$.

7. Association vs. Causality

Concept	Meaning
Descriptive Question	Asks about patterns in data, e.g., how spending

	varies with income
Research Question	Asks about causal effect, e.g., effect of attendance on learning
Observational Data	Variables are naturally observed; lurking variables may exist
Experimental Data	Researcher randomly assigns x values when possible
Endogeneity	x is associated with unobserved factors that also affect y
Exogeneity	x is not associated with factors that also affect y
Core sentence: Correlation does not necessarily imply causation. A strong correlation may be driven by omitted variables, reverse causality, or selection.	

Practice 7 - Lurking Variable

Educational data show that taller elementary school students have higher reading scores. What is the likely lurking variable?

Answer: Age is a lurking variable because it affects both height and reading ability.

Practice 8 - Endogeneity

Researchers find a positive correlation between long-term PM2.5 exposure and dementia rates. Why should we be careful before making a causal claim?

Answer: PM2.5 exposure may be endogenous. People do not randomly choose where to live; income, education, health, smoking, and neighborhood conditions may affect both pollution exposure and dementia risk.

8. Integrated Practice Bank

Practice 9 - Strongest Positive Correlation

Given correlations $r_{wx}=0.3251$, $r_{wy}=0.1711$, $r_{wz}=0.6758$, $r_{xy}=0.1259$, $r_{xz}=0.4332$, $r_{yz}=0.1873$, which pair has the strongest positive correlation?

Answer: The strongest positive correlation is between w and z because 0.6758 is the largest positive correlation.

Practice 10 - Variance-Covariance Matrix

If $s_x = 3$, $s_y = 4$, and $r_{xy} = 0.5$, find s_{xy} . What are the diagonal elements of the variance-covariance matrix?

Answer: $s_{xy} = r_{xy} s_x s_y = 0.5(3)(4) = 6$. Diagonal elements are variances: $s_x^2 = 9$ and $s_y^2 = 16$. Matrix: $\begin{bmatrix} 9 & 6 \\ 6 & 16 \end{bmatrix}$.

Practice 11 - Panel Data

Pollution data are observed for 103 water stations across 5 years. What type of data is this, and what is the unit of observation?

Answer: This is panel data. The unit of observation is a water station in a particular year. There are $103 \times 5 = 515$ station-year observations before accounting for missing values.

9. Lecture 2 Summary

- Use scatterplots before computing correlation.
- Covariance gives direction but its magnitude depends on units.
- Correlation measures direction and strength of a linear relationship.
- r close to zero does not rule out nonlinear relationships.
- Pearson r is sensitive to outliers; Spearman correlation uses ranks.
- Association is not causation; check endogeneity and lurking variables.

Lecture 2 takeaway: Use scatterplots to see the pattern, covariance to identify direction, correlation to measure linear strength, and causality concepts to avoid over-interpreting association.